

What is a multimeter?

According to the traditional and most basic classification, the multimeters are categorised in two types—the analogue and digital. Analogue multimeters are typically moving-coil instruments that measure values continuously and display the reading on a calibrated scale using a pointer.

The digital multimeter (DMM) displays measured values at discrete intervals of time on an LCD screen. Compared to analogue multimeters, DMMs commonly have some additional features. They are used as a continuity tester, that is, they produce a beep sound when two components are electrically connected. This feature can be used to check whether the components are properly connected, are there any breaks in the circuit or line, or to detect a short circuit.

The forward voltage of diodes can also be measured with DMMs. Most DMMs can also measure current gain, other parameters and even identify the leads of transistors.

In addition to these features, the users can attach various sensors to some DMMs to measure light and sound level, relative humidity, acidity or alkalinity and wind speed.

How to measure cost for your test system?

Price point of the tool is very important as these tools have cutting-edge technology and, as newer devices enter the market, the product differentiation is a must.

The cost is generally not measured in terms of the individual price points of a DMM or oscilloscope or waveform generator, but as the overall cost of ownership of the entire system and the long-term return of investment users get. It is highly dependent on the application—the number of instruments or the number of measurements a user needs to make. Considering the complexity of systems we are looking at today, the overall opinion is that the systems built on a modular platform definitely provide higher returns than the traditional instruments.

The other aspect is the cost of developing equipment within a system and its long-term maintenance. This is usually achieved with highly integrated and efficient software.

and simplify user interface. It is important that the software controlling these instruments is highly integrated and efficient.

Integrated DMMs

Nowadays, people expect more measurement capabilities than the traditional voltage, current and resistance measurements that were typically performed by a multimeter. Research is ongoing for including more measurement capabilities, quick and better integration and development of the intuitive abilities of the application. This R&D is focussing on the inclusion of additional measurements by integrating DMM with other sensors, measuring devices and switches.

Currently, more devices such as voltage and current amplifiers, source measure units (SMUs) and digitisers or oscilloscopes are combined with the typical DMM functionality.

The integration of several instruments into a single module helps in

developing 'managed test systems' that allow users to combine traditional multimeter functions with other elements/functions such as system monitoring, diagnostics, data logging, configuration and calibration.

Now it is no longer a stand-alone DMM, instead it has additional capabilities of attached elements that are combined in a single instrument. Such DMMs allow the users to take readings from multiple points and make more enhanced digital measurements in order to characterise their systems much more accurately, thus increasing customer efficiency.

So what we can expect in the DMM space is the ability to create, design and develop very high-end specification instruments.

The need for such integrated products depends on the application field, the variety of the user's job roles and their affordability. Now there is a provision for the users to buy a base instrument with limited and affordable

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